Soham Patel

Software Engineer — Physical AI, Simulation & Data Systems

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PROFILE

Software engineer with about 2.5 years of production experience in healthcare data and payment systems, now studying Computer Engineering at Paderborn University with a focus on Physical AI. My project work sits between software and hardware: a MuJoCo digital twin of a fischertechnik factory cell built from its PLC step tables and verified against 56 checks; LIDAR SLAM with occupancy-grid mapping and pure-pursuit control, ported from a Python Webots controller into the browser; YOLOv8 exported to ONNX for on-device inference; diffusion-based anomaly detection on brain MRI, with the weak result stated plainly. M.Sc. Information Technology as Gold Medalist, Rank 1 of cohort (9.55/10).

PROFESSIONAL EXPERIENCE

03/2025 – 08/2025
Part-time

Teaching Assistant — Decision Modelling & Data Analysis
OTH Amberg-Weiden · Weiden, Germany · Supervisor: Prof. Dr. Simon J. Preis

- Designed and delivered the applied data-analysis strand of a Master's module to a cohort of 30: SQL, statistical modelling, classification, regression and model evaluation.
- Built and maintained the KNIME reference workflows used across the module, covering data cleaning, feature engineering and evaluation on real industrial datasets.

02/2023 – 02/2024
Full-time

Software Engineer — Data Systems
Altera Digital Health (Allscripts India LLP) · Pune, India / Remote

- Built ETL pipelines on Microsoft Azure for healthcare audit-log data; data retrieval became about 30% faster.
- Implemented the encryption, access logging and retention rules required for health data, enforced in code rather than by convention.
- Refactored legacy C#/.NET services and introduced pytest checks that run before each deployment.

C# · .NET · Microsoft Azure · Blob Storage · T-SQL / MSSQL · Azure DevOps · pytest · Git

08/2022 – 01/2023
Full-time

Python Backend Engineer
Electrum IT Solutions · Vadodara, India

- Built FastAPI and Django services for a payments platform handling about 2,000 transactions a day.
- Implemented JWT authentication and role-based access control across the service boundary.
- Introduced centralised logging for failed payments, which made debugging about 25% faster.

Python 3 · FastAPI · Django · PostgreSQL · JWT · REST · Linux

12/2021 – 04/2022
Internship

Full-Stack Engineering Intern
Promact Infotech Pvt. Ltd. · Vadodara, India

- Delivered an expense-management product end to end: Angular 8 front end, ASP.NET Core 5 API, 9-table normalised MSSQL schema, JWT auth with role-based access, WebSocket live balances.
- Designed a graph-theory debt-simplification algorithm reducing the transfers needed to settle a group by up to 33%.

Angular 8 · ASP.NET Core · MSSQL · WebSocket · JWT · Agile

EDUCATION

Current

M.Sc. Computer Engineering
Paderborn University · Paderborn, Germany

- In progress. Focus on Physical AI: robotics, embedded systems and the software that connects sensing to action.

10/2024 – 08/2026

Graduate Coursework — Artificial Intelligence for Industrial Applications
Ostbayerische Technische Hochschule Amberg-Weiden · Amberg, Germany

- 45 ECTS · GPA 2.0. Digital twins and Industry 4.0, computer vision, generative models, autonomous systems (Embedded Intelligence graded 1.3).

08/2020 – 05/2022 **M.Sc. Information Technology**
Charutar Vidya Mandal University · Gujarat, India
- CGPA 9.55/10. **Gold Medalist, Rank 1** of a graduating class of approximately 120.
- Software engineering, databases and distributed systems; thesis in applied machine learning.

06/2017 – 05/2020 **B.Sc. Computer Science**
Sardar Patel University · Gujarat, India
- CGPA 8.5/10. Algorithms and data structures, operating systems, networks.

SELECTED PROJECTS

10/2025 – 03/2026 **Physics Digital Twin — fischertechnik Training Factory**

- MuJoCo rigid-body simulation of all four stations of the fischertechnik 536634 cell, built from the manufacturer's PLC step tables, TON timer literals and datasheets rather than from photographs; verified against 56 PLC-derived checks with 1.2 mm RMS prediction.
- Earlier versions: a browser 3D twin driven by the real PLC process image (9 DI / 14 DO), and an MQTT/InfluxDB maintenance dashboard (85% failure-prediction accuracy). Live on Google Cloud Run.

MuJoCo · Python · Siemens SCL · MQTT · InfluxDB · Three.js · Docker

2026 **Unsupervised Brain-MRI Lesion Segmentation (BraTS-PEDs)**

- Patch diffusion in pixel space: a DDPM trained on healthy 96×96 patches only; at inference each patch is partially noised and reconstructed, and the residual localises the lesion, refined by a dense-CRF guided by T1c contrast.
- **Result: DICE 0.076 · AUROC 0.713** — the masks do not yet segment usefully; reported as a baseline. The earlier latent-diffusion version underperformed and is documented as a negative result.

PyTorch · DDPM / DDIM · dense-CRF · HuggingFace · CUDA · Docker

2025 **Autonomous Maze Navigation — LIDAR SLAM**

- Log-odds occupancy-grid mapping, frontier exploration, A* planning on an inflated obstacle radius, pure-pursuit steering; five Webots maze worlds, ported from the Python controller into the browser, where the port is covered by unit tests.

Python · Webots · NumPy · Three.js · TypeScript

2025 **Real-Time Scene Understanding**

- YOLOv8n-seg exported to ONNX and run under ONNX Runtime Web, so camera frames never leave the visitor's device; paired with a self-hosted vision-language model for narration, with a detector-only fallback when none is reachable.

YOLOv8 · ONNX Runtime Web · TypeScript · FastAPI · Docker

TECHNICAL SKILLS

Languages	Python 3 · C# / .NET · TypeScript · SQL (T-SQL, MSSQL, PostgreSQL, MySQL)
Backend & Cloud	FastAPI · Django · ASP.NET Core · REST · JWT · Microsoft Azure · Docker · CI/CD · Git · Linux · pytest
ML & Vision	PyTorch · Scikit-Learn · XGBoost · HuggingFace · YOLOv8 · ONNX Runtime · Diffusion models (DDPM/DDIM) · Anomaly detection
Industrial & Data	Digital twins · MQTT · OPC-UA · PLC process images · InfluxDB · ETL pipelines · SCADA time-series · KNIME
Robotics & 3D	Webots · SLAM (occupancy grids) · A* / pure pursuit · MuJoCo · Three.js

AWARDS, LANGUAGES & ADDITIONAL

2025 **Grade 1.0 (highest distinction)** — keynote, OTH AI Research Conference: *Brownian Bridge Diffusion Models for 3D Shape-to-Image Medical Image Synthesis*.

2022 **Gold Medal & University Rank 1** — M.Sc. IT, CVM University (CGPA 9.55/10).

Languages English C1 (IELTS 6.5) · German A2, in active study toward B1 · Gujarati native · Hindi fluent.

Department student representative (two years) · Football: university team captain.

Work authorisation Valid German residence permit · EU Blue Card eligible via a recognised M.Sc. · no sponsorship required · available from September 2026.